

Zengjun Guo

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Research Interests

- **Urban Resilience & Disaster Risk Reduction:** resilience assessment of critical infrastructure systems and their interactions with human responses under compound hazards.
- **Climate Risk Modeling & Adaptation:** probabilistic simulation of climate extremes and scenario-based projections of their impacts across regions and sectors.
- **Urban Analytics & GeoAI:** large-scale spatiotemporal data and geospatial machine learning to characterize human mobility and network dynamics during disasters.

Education

Peking University, China

Sep 2024–Present

M.Sc. in Territorial Spatial Planning

GPA: 3.84/4.0

Nankai University, China

Sep 2020–Jun 2024

B.Mgmt. in Urban Management

GPA: 3.85/4.0 (ranked 1st)

University of Glasgow, UK

Sep 2022–Jun 2023

Exchange Student in Urban Studies

GPA: 18.3/22.0 (First-Class)

Research Experience

School of Urban Planning and Design, Shenzhen Graduate School, Peking University

Graduate Researcher

Shenzhen, China / Sep 2024–Present

Advisor: Dr. Junqing Tang

- Generated synthetic commuting OD matrices with GLODGen and tropical cyclone tracks from CMIP6 forcing; coupled C15–TCR wind–rain modeling, asset valuation, and network assignment to assess global urban road damage and mobility disruption.
- Assessed older-adult exposure and healthcare mismatches under tropical cyclones in 52 Chinese coastal cities; projected annual hospital asset losses to more than double under SSP5–8.5, with climate change amplifying aging-driven exposure growth by 19.7–32.8%.
- Reconstructed Mangkhut’s wind–rain evolution with ER11 and GMCP and processed 2.5 billion mobile-phone records to reveal coast–inland resilience gaps and future inland vulnerability shifts.

Key Laboratory of Earth Surface System and Human-Earth Relations (MNR)

Research Group Member

Shenzhen, China / Oct 2023–Present

Advisors: Dr. Junqing Tang and Prof. Pengjun Zhao

- Proposed a human–environment–disaster coupling theory, introduced the marginal constraint effect to define disaster risk as an endogenous constraint, and validated the theory using evidence from the 2021 Zhengzhou “7 · 20” flood.
- Mapped flood-induced shelter demand–supply mismatches across 21 cities with GHS-SMOD and developed an interactive spatial decision system for targeted emergency resource allocation.
- Refined and applied a DBN–ARIMA framework across nine Pearl River Delta cities, identifying four 2015–2021 resilience trajectories and projecting a widening Guangzhou–Shenzhen-centered core–periphery pattern through 2026.

Selected Publications & Manuscripts

Articles and Manuscripts

1. **Guo, Z.**, An, M., Ye, L., Zhao, P., Tang, J.* 2026. Risk pattern and spatiotemporal evolution of healthcare-service supply in China’s coastal cities under tropical-cyclone wind disasters. **Acta Geographica Sinica** (under review). (In Chinese)
2. Tang, J., **Guo, Z.**, Lyu, W., Zhao, P.* 2026. The geographical connotation of resilient-city research: human–environment relationships under disaster-risk constraints. **Progress in Geography**, 45(4), 683–694. (In Chinese)

3. Tang, J., Lyu, W., Zhao, P.^{*}, Huang, F., Li, J., Fan, C., Lu, Q., Wang, J., Ye, L., **Guo, Z.** 2026. Human mobility dynamics for cost-effective flood emergency resource allocation. **Nature Sustainability**. doi:10.1038/s41893-026-01886-9.
4. **Guo, Z.**, Tang, J.^{*}, Huang, F., Zhao, P.^{*} Shifting hotspots of compound tropical cyclone risk to global urban road systems in a warming climate. *In preparation*.
5. Huang, F.[†], **Guo, Z.**[†], Lyu, W., Zhao, P., Tang, J.^{*} Assessing the 15-minute city during extreme urban floods: Evidence from Shenzhen and Zhengzhou, China. *In preparation*.
6. Wang, J., Tang, J.^{*}, **Guo, Z.**, Zhao, P. 2026. Assessing spatial-temporal resilience of road transportation against extreme typhoon hazards in coastal cities: a case study in Shenzhen, China. **Journal of Transport Geography** (major revision).
7. Wang, T., Xu, T.Z.^{*}, Feng, H., **Guo, Z.**, Poo, M.C.-P., Tang, J., Ng, A.K.Y. 2026. Enhancing port resilience to typhoon disasters: a framework for recovery optimization and shipping network impact assessment. **Transport Policy**, 180, 104029.
8. Tang, J., Han, S., **Guo, Z.**, Ye, L., Wang, H.^{*} 2026. Urban resilience assessment of the Guangdong–Hong Kong–Macao Greater Bay Area based on a dynamic Bayesian network model. **Journal of Natural Disasters** (minor revision). (In Chinese)

Book chapter

9. **Guo, Z.**, Tang, J. 2026. Japan’s comprehensive efforts to mitigate damage from earthquake hazards (Ch. 11). In J. Tang, Q. Lu, P. Zhao, I. Brilakis, T. Broyd & L. Chen (Eds.), **Resilient Cities under Global Climate Change: From Principles to Applications**. Edward Elgar Publishing.

Conference Presentations

1. “The Geographic Connotation of Resilient-City Research: An Exploration of Disaster-Risk-Constrained Human–Earth Relationships.” Oral presentation at the 13th International Conference on Population Geographies (ICPG), Nanjing, China, Jun 2026.
2. “Human–Environment Relationships under Disaster-Risk Constraints: A Preliminary Theoretical Exploration of the Geographic Connotation of Resilient-City Research.” Oral presentation at the 2026 Spring Annual Meeting of the Geographical Society of China, Chongqing, China, Apr 2026.

Academic Service

1. Committee Member, Sub-Society for Postgraduate Students, Geographical Society of China, 2025–Present.

Honors & Awards

1. 2026 First Prize, 10th Chengyuan Cup Planning Decision Support Model Design Competition.
2. 2025, 2026 Second Prize, Peking University Academic Science & Technology Competition.
3. 2025 Third Prize, Esri Cup China University GIS Software Development Contest.
4. 2024 National Second Prize, Undergraduate Innovation & Entrepreneurship Training Program.
5. 2024 Outstanding Undergraduate Thesis, Nankai University.
6. 2024 Outstanding Graduate, Nankai University.
7. 2024 Huadan Scholarship in Political Science, Nankai University.
8. 2021, 2023 First-Class Scholarship, Nankai University.
9. 2022 Tianjin Municipal Government Scholarship, Tianjin Municipal Education Commission.

Technical Skills

Programming & web development: Python (NumPy, pandas, GeoPandas, Rasterio, xarray, scikit-learn); JavaScript; HTML/CSS.

GIS & modeling software: ArcGIS; GeoScene Pro; QGIS; CLIMADA; GeNIe Modeler.

Development & typesetting: Git/GitHub; LaTeX.